Predavanja

ustni izpit: 40%

predstaviteev: 20% (20 minut po 3 osebe)

Vaje

Projekt 40%

Logic in computer science

- Correctness of software/hardware/systems
  (watch Xavier Leroy: In search of sofware prfection)
- Developing proof asistants

Propositional logic (natural deduction proof system)
Simple type theory : Curry-Howard correspondance
(propositions as types)

Propositional logic

propositional formulas ϕ, ψ, θ

propositional variables p

atomic formula:

$$\phi ::= p \mid \psi \wedge \theta \mid \psi \vee \theta \mid \psi \to \theta \mid \neg\theta \mid \bot$$

TAVTOLOGIJA - validna formula (vedno res)

Natural deduction: proof system. Using formal rules we prove a formula ϕ

We can prove formula $\Leftrightarrow$ it is a tavtology

Primer: Proof of $p \wedge q \to p \vee r$

1. Assume $p \wedge q$ true (assumption)
2. p ($\wedge$elimination$_1$)
3. $p \vee r$ ($\vee$introduction$_1$)
4. $p \wedge q \to p \vee r$ ($\to$ introduction)

Za konstruktivno logiko (Brez LEM

$\wedge$

$$\frac{\phi \quad \psi}{\phi \wedge \psi}\wedge i \qquad \frac{\phi \wedge \psi}{\phi}\wedge e_1 \qquad \frac{\phi \wedge \psi}{\psi}\wedge e_2$$

premises

conclusion

$\vee$

$$\frac{\phi}{\phi \vee \psi}\vee i_1 \qquad \frac{\psi}{\phi \vee \psi}\vee i_2 \qquad \frac{\phi \vee \psi \quad \boxed{\begin{matrix}\phi\\ \vdots\\ \theta\end{matrix}} \quad \boxed{\begin{matrix}\psi\\ \vdots\\ \theta\end{matrix}}}{\theta}\vee e$$

$\rightarrow$

$$\frac{\boxed{\begin{matrix}\phi\\ \vdots\\ \psi\end{matrix}}}{\phi \rightarrow \psi}$$

box discharges the assumption ϕ

$$\frac{\phi \rightarrow \psi \quad \phi}{\psi}\rightarrow e$$

modus ponens

$\neg$

$\neg\phi \equiv \phi \rightarrow \bot$

$$\frac{\boxed{\begin{matrix}\phi\\ \vdots\\ \bot\end{matrix}}}{\neg\phi}\neg i \qquad \frac{\neg\phi \quad \phi}{\bot}\neg e$$

$\bot$

$$\frac{\bot}{\phi}\bot e$$

NO FORMUA IN THE BOX gets used outside the box.

Primer $p \wedge (q \vee r) \rightarrow (p \wedge q) \vee (p \wedge r)$

1. $p \wedge (q \vee r)$ assumption
2. p
3. $q \vee r$
4. q assumption
5. $p \wedge q$
6. $(p \wedge q) \vee (p \wedge r)$
7. r assumption
8. $p \wedge r$
9. $(p \wedge q) \vee (p \wedge r)$ $\vee e$ (3) (4-6) (7-9)
10. $(p \wedge q) \vee (p \wedge r)$ $\rightarrow i$ (1-10)

In classical logic

$$\frac{\boxed{\begin{matrix}\neg\phi\\ \vdots\\ \bot\end{matrix}}}{\phi}raa$$

Reductio ad absurdum

DN: prove LEM with raa

$$\frac{}{\phi \vee \neg\phi} \qquad \frac{\dfrac{\dfrac{\dfrac{}{\neg\phi}}{\bot}}{\phi}}{\phi \vee \neg\phi} \qquad \frac{\dfrac{}{\neg\phi}}{\phi \vee \neg\phi}$$

$$\phi \vee \neg\phi$$

A proof of formula ϕ from assumptions Γ is given by sequence of formulas following the rules such that all open assumptions in the proof are from Γ

notation: $\vdash \phi$

Theorem: $\vdash \phi$ (including raa) $\overset{\text{soundnes}}{\underset{\text{completness}}{\Longleftrightarrow}}$ ϕ is tautology

$\Leftrightarrow$ ϕ is true under any assigment of truth validnes to propositional variables that make all formulas in Γ true

Formulas $\phi ::= p \mid \phi \wedge \phi \mid \phi \vee \phi \mid \phi \to \phi \mid \bot$

Types $A ::= \alpha \mid A \times B \mid A + B \mid A \to B \mid$

$$t : A$$

term type

$$\frac{s : A \quad t : B}{(s, t) : A \times B} \qquad \frac{t : A \times B}{pr_1(t) : A}\times e_1 \qquad \frac{t : A \times B}{pr_2(t) : B}$$

variable

$$\frac{\begin{array}{c} x : A \\ \vdots \\ t : B \end{array}}{\lambda x : A . t : A \to B} \to i \qquad \frac{t : A \to B \quad u : A}{t u : B} \to e$$

term

$$\frac{t : A}{in_1(t) : A + B} \qquad \frac{t : B}{in_2(t) : A + B}$$

$$\frac{s : A + B \quad \boxed{\begin{array}{c} x : A \\ \vdots \\ t : C \end{array}} \quad \boxed{\begin{array}{c} y : B \\ \vdots \\ u : C \end{array}}}{\text{case } s \, (in_1(x : A) \mapsto t \mid in_2(y : B) \mapsto u) : C}$$

Primer:

A derived term of type $A \times B \to B \times A$

1.	$x : A \times B$	assumption	
2.	$pr_1(x) : A$	$\times e_1$	(1)
3.	$pr_2(x) : B$	$\times e_2$	(1)
4.	$(pr_2(x), pr_1(x)) : B \times A$	$\times i$	(3) (2)
5.	$\lambda x : A \times B . (pr_2(x), pr_1(x)) : A \times B \to B \times A$	$\to i$	(1-4)

tega ne potrebujemo

For every type we have introduction rules, eliminatio rules and computation rules

computation rule for rewrite rule (longer arrow)

$\times$: $pr_1(s,t) \longrightarrow s$

$pr_2(s,t) \longrightarrow t$

β-reduction

substitute u for x in t

$\rightarrow$: $(\lambda x{:}A.t)u \longrightarrow t[x:=u]$

(before doing this first rename bound variable in t that are distinct from variables in u)

$+$: $\text{case}\ (in_1(s))\ (in_1(x) \mapsto t \mid in_2(y) \mapsto u) \longrightarrow t[x:=s]$

0 : $\dfrac{t:0}{\text{empty}(t):A}$

DM: Find canonical term of type

$A \times (B+C) \longrightarrow A\times B + A \times C$

Propositional logic ~ Simple Type theory

Predicate logic ~ Dependent type theory
(first order logic

Propositional logic formulas

$$\phi ::= p \mid \phi \wedge \psi \mid \phi \vee \psi \mid \phi \to \psi \mid \bot$$

arity ≥ 0

Predicate logic extends prop. logic with <u>predicates</u> expressing relations, and <u>constants and function</u> symbols expressing elements

arity ≥ 0

Terms express elements

$$t ::= x \mid c \mid f(t_1, \ldots, t_n)$$

constant

function symbol of arity n

Formulas express propositions

$$\phi ::= P(t_1, \ldots, t_n) \mid \phi \wedge \psi \mid \phi \vee \psi \mid \phi \to \psi \mid \bot \mid \forall x \phi \mid \exists x. \phi$$

Example: a language for expressing property of integers in predicate logic

Constants: $0, 1$

function symbols $+$, $\cdot$ of arity 2

example terms z, $x+y$, $x\cdot(y+(z+1))$

predicates $=$, $\leq$

example formulas $x \leq y$, $\exists y \exists z \exists w \exists v.\ x = yy + z\cdot z + w\cdot w + v\cdot v$

free variable

We could also define $\leq$ as

$$x \leq y \Leftrightarrow \exists x. \exists y. \exists v. \exists u.\ y = x + z\cdot z + w\cdot w + v\cdot v + u\cdot u$$

(vsa naravna števila sa vsote štirih kvadratov)

$x+1=0 \quad (x=-1)$

We will interpreate predicate logic in type theory

Prop ... type of propositions

$$\frac{\phi : Prop \quad \psi : Prop}{\phi \wedge \psi : Prop} \wedge \text{form} \qquad \frac{\phi : Prop \quad \psi : Prop}{\phi \vee \psi : Prop} \vee \text{form.}$$

$$\frac{\phi : Prop \quad \psi : Prop}{\phi \to \psi : Prop} \to \text{formation} \qquad \frac{}{\bot : Prop}$$

(1) $p : Prop$ assumption

(2) $q : Prop$ assumption

} open assumption are called a context

(3) $p \wedge q : Prop$ $\wedge$ formation (1) (2)

(4) $q \wedge p : Prop$ $\wedge$ formation (2) (1)

(5) $p \wedge q \to q \wedge p : Prop$ $\to$ formation (3) (4)

$$p : Prop, q : Prop \vdash p \wedge q \to q \wedge p : Prop$$

For predicate logic we also neen quantifiers

Predicate logic

$$\frac{\boxed{\begin{array}{c} x:A \\ \Phi:Prop \end{array}}}{\forall x:A.\,\Phi:Prop}\ \forall\text{-formation} \qquad \frac{\boxed{\begin{array}{c} x:A \\ \Phi:Prop \end{array}}}{\exists x:A.\,\Phi:Prop}\ \exists\text{-formation}$$

(formation, not introduction

①	$p: A \to Prop$	assumption
②	$x:A$	assumption
③	$p\,x : Prop$	$\to e$
④	$\forall x:A.\,px : Prop$	$\forall$-formation (2) (3)

$$p: A \to Prop \vdash \forall x:A.\,px : Prop$$

(Insted of formation rules, we can include conectives and quantifiers as typed constants

$\to, \vee, \wedge : Prop \to prop \to Prop$

$\exists, \forall : (A \to Prop) \to Prop$)

iz Primera: $\forall(x:A.\,p(x))$

Example proof of

$$p : A \to Prop,\ q : A \to Prop \vdash (\forall x : A.\, px) \to (\forall x : A.\, qx) \to (\forall x : A.\, px \wedge qx)$$

3 judgments forms

$t : A$

$\Phi : Prop$ "Φ is prop."

Φ "Φ is true"

1	$p : A \to Prop$	assump.
2	$q : A \to Prop$	assump.
3	$\forall x : A.\, px$	assump
4	$\forall y : A.\, qy$	assump
5	$z : A$	assum
6	$p\,z$	$\forall E$ (3)
7	$q\,z$	$\forall E$ (4)
8	$p\,z \wedge q\,z$	$\wedge I$ (6) (7)
9	$\forall z : A.\, pz \wedge qz$	$\forall I$ (5)-(8)
10	$\forall y : A.\, qy \to \forall z : A.\, pz \wedge qz$	$\to$ (4)-(9)
11	$\forall x : A.\, px \to (\ldots) \to (\ldots)$	$\to$ (3)-(10)

z se ne sme pojaviti v kontekstu preg

$$\frac{\boxed{\begin{array}{c} z : A \\ \vdots \\ \Phi \end{array}}}{\forall z : A.\, \Phi}\ \forall I$$

$$\frac{\forall x : A.\, \Phi \qquad t : A}{\Phi[x := t]}\ \forall E$$

Simply we replace the grammar for every type with a <u>universe</u> Type and formation rules for types

We had: $A ::= \alpha \mid A \times A \mid A + A \mid A \to A \mid 0$

$$\frac{A : \mathrm{Type} \quad B : \mathrm{Type}}{\begin{array}{c} A + B : \mathrm{Type} \\ A \times B : \mathrm{Type} \\ A \to B : \mathrm{Type} \end{array}}$$

Key idea of dependent type theory is types depend on the context

$\mathbb{N}$ set of natural numbers

List $\mathbb{N}$ set of lists (=finite sequences) of nat. nums

Vector $\mathbb{N}$ n set of lists of n nat. numbers of length n

Dependant product

construction $\prod_{n:\mathbb{N}} \text{Vector}\ \mathbb{N}\ n$ | function f that map $n:\mathbb{N}$ and return a value from the set $\text{Vector}\ \mathbb{N}\ n$

examples

$$n \mapsto \overbrace{0 \ldots 0}^{n}$$

$$n \mapsto \underbrace{1, 2, \ldots, n}_{n}$$

$$\frac{\boxed{\begin{array}{l} x:A \\ \vdots \\ B:\text{Type} \end{array}}}{\Pi x:A.\ B:\text{Type}}\ \Pi\text{-formation}$$

type Nat

$$\frac{}{nat : Type}$$ formation rule

$$\frac{}{0 : nat} \qquad \frac{t : nat}{succ\ t : nat}$$ introduction rules

alternative

$succ : nat \to nat$

Elimination and computation rules latter

$$\frac{t : nat \qquad P : nat \to Prop \qquad _ : P0 \qquad \boxed{\begin{array}{c} x : nat \\ \vdots \\ P(succ\ x) \end{array}}}{Pt}$$ Introduction rule

Instead of rule based formulations we can assume rules:

introduction: $0 : nat \qquad succ : nat \to nat$

elimination:

$$\forall P : nat \to Prop \left(P0 \to (\forall x : nat.\ Px \to P(succ\ x)) \to \forall x : nat.\ Px \right)$$

induction

Iz indukcije v matematiki izrazimo recursion theorem

if v have A set and $a \in A$ and $A \to A$ then there exist unique function $\Gamma : \mathbb{N} \to A$ da $\Gamma(0) = a$ $\Gamma(n+1) = f(\Gamma(n))$

Recursion principle

$$R_{nat} : \Pi_{P : nat \to Type} (P0 \to (\Pi_{x:nat}. Px \to P(succ\, x)) \to \Pi x : nat\ Px)$$

recursor

$$R_{nat}\ P\ b\ f\ : \Pi_{x:nat} Px$$

computation rule:

$$R_{nat}\ P\ b\ f\ 0 \longrightarrow b$$

$$R_{nat}\ P\ b\ f\ (succ\ n) \longrightarrow fn(R_{nat}\ P\ b\ f) : P\,succ(n)$$

Example:

addition $\quad add : Nat \to Nat \to Nat$

$$add := \lambda m{:}nat.\ \lambda n{:}nat.$$
$$R_{nat}\ (\lambda_{:}nat.\ Nat)\ m\ (\lambda_{:}nat.\ \lambda x{:}nat.\ succ\ x)\ n$$

$$add\ \overbrace{(succ\ 0)}^{m}\ \overbrace{(succ\ 0)}^{n} \longrightarrow^*$$
$$R_{nat}\ (_{:}nat.\ nat)\ succ\ 0\ \lambda_{:}nat.\ \lambda x{:}nat.\ succ\ x)\ succ\ n$$
$$\longrightarrow (\lambda_{:}nat.\ \lambda x{:}nat.\ succ\ x)\ 0\ ((R_{nat}\ \lambda_{:}nat.\ nat)$$
$$(succ\ 0)\ (\lambda_{:}nat.\ \lambda x{:}nat.\ succ\ x)\ 0\ ))$$
$$\longrightarrow^* succ\ (R_{nat}(\lambda_{:}nat.\ nat)\ (succ\ 0)$$
$$(\lambda_{:}nat.\ \lambda x{:}nat.\ succ\ x)\ 0)$$
$$\longrightarrow succ\ (succ\ 0) = 2 \qquad \ddot{\smile}$$

type List

$$\frac{A \text{ type}}{\text{list}A : \text{Type}}$$ formation

$$\frac{}{\text{nil} : \text{List}\,A} \qquad \frac{a : A \qquad s : \text{List}\,A}{\text{cons}\ s\ a : \text{list}A}$$

nil : list A

cons : $A \to \text{List}\,A \to \text{List}\,A$. alternative

elimination, computation rules

Vector type

$$\frac{A : Type \qquad n : nat}{vector\ A\ n\ Type}$$

$$\frac{}{nil : vector\ A\ 0} \qquad \frac{a : A \qquad as : vector\ A\ n}{cons^{(n)}\ a\ as : vector\ A\ (succ\ n)}$$

alternative constructs:

nil : vector A 0

cons :

$$\prod_{n:nat} A \to Vector\ A\ n \to Vector\ A\ (succ\ n)$$

Vector is dependent type, because it's depends on the value of n

$$Rvector : \prod_{P : \prod_{n:nat} Vector\ A\ n \to Type} P\ 0\ nil \to \prod_{a:A} \prod_{n:nat} \prod_{as:vector\ A\ n} P\ n\ as \to P\ (sup\ n)\ (cons\ n\ a\ as)) \to \prod_{n:nat} \prod_{as:vector\ A\ n} P\ n\ as$$

Computation rules

$$P : \prod_{n:nat} (vector\ A\ n \to Type)$$

$b : P\ 0\ nil$

$f :$

$$Rec_{vector\ A}\ P\ b\ f : \prod_{n:nat} \prod_{as:vec\ A\ n} P\ n\ as$$

$$Rec_{vec}\ P\ b\ f\ 0\ nil \longrightarrow b$$

$$Rec_{vec}\ P\ b\ f\ (succ\ n)\ (cons\ n$$

??

?

nat, List, vectors so induktivní typi

Dependent product type

$$\frac{A \text{ type} \qquad \boxed{\begin{array}{c} x:A \\ \vdots \\ B \text{ type} \end{array}}}{\Pi_{x:A} B : \text{Type}}$$

intuitively: $\Pi_{x:A} B$ type of functions that maps $x:A$ to type B

introduction

$$\frac{\boxed{\begin{array}{c} x:A \\ \vdots \\ t:B \end{array}}}{\lambda x:A.t : \Pi_{x:A} B}$$

elimination

$$\frac{s : \Pi_{x:A} B \qquad t:A}{s\,t : B[t/x]}$$

Ko imamo odvisni produkt, ne potrebujemo $\rightarrow$

$A \rightarrow B := \prod_{x:A} B$, kjer B ne vsebuje x

computation rule: $(\lambda x:A.t)u \rightarrow t[u/x]$

β-reduction

Dependent sum type

$$\frac{A : \mathrm{Type} \qquad \boxed{\begin{array}{c} x : A \\ \vdots \\ B : \mathrm{Type} \end{array}}}{\sum_{x:A} B : \mathrm{Type}}$$

$$\frac{s : A \qquad t : B[s/x]}{(s,t) : \sum_{x:A} B}$$

$$\frac{t : \sum_{x:A} B}{pr_1(t) : A} \qquad \frac{t : \sum_{x:A} B}{pr_2(t) : B[pr_1(s)/x]}$$

$$\left(A \times B := \sum_{x:A} B \right)$$

Types and their elements

0	no elements
$A \times B$	pairs (a,b), $a:A$ $b:B$
$A+B$	$in_1(a)$; $a:A$ or $in_2(b)$; $b:B$
$A \to B$	functions from A to B
$\prod_{a:A} B$	functions from $a:A \to B[a/x]$
$\sum_{a:A} B$	pairs (a,b) where $a:A$ $b:B[a/x]$

The Brower-Heyting-Kolmogorov interpretation of logic

$\bot$ has no proof

proof of $\Phi \wedge \Psi$ is pair (a,b) where a is proof of Φ and b is proof of Ψ

propositions as subterminal types

if $\Phi : Prop$ then $\Phi : Type$

($Prop$ is a subuniverse of $Type$

$Prop$ is closed under Π-type

$$\left(\frac{A\ type \qquad \boxed{\Phi\ Prop}^{x:A}}{\Pi a:A\ B : Prop} \right)$$

if Φ is $Prop$ then Φ is subterminal

– if $a, b : \Phi$ then $a = b$ (equality next week)

Also: $Prop : Type$ – impredicativity

We can take Π as the only constructor

Scott-Prawitz encoding of intuitionistic logic

$\bot := \Pi_{p:prop}\, p$

"for all propositions p, p is true"

$$\frac{a : \Pi_{p:prop} \qquad \Phi : prop}{a\,\Phi : \Phi} \qquad \text{taej} \qquad \frac{\bot \qquad \Phi\ prop}{\Phi}$$

$\Phi \wedge \psi \qquad \Pi_{p:prop}\, (\Phi \to \psi \to p) \to p$

$\Phi \vee \psi \qquad \Pi_{p:prop}\, (\Phi \to p) \to (\psi \to p) \to p$

$\Phi \to \psi$

$\forall x{:}A.\, \Phi \qquad \Pi_{x:A}\, \Phi$

$\exists x{:}A\, \Phi \qquad \Pi_{p:prop}\, (\Pi_{x:A}\, (\Phi \to p)) \to p$

Basic properties

$A : \mathrm{Type}$

$t : A$ — needs proof that $A : \mathrm{Type}$

$s \equiv t : A$ s, t are judgmentaly equal

Derived judgment forms

$\Phi : \mathrm{Prop}$ ... instance of $t : A$

$t : \Phi$

Φ means $_\ \Phi$ (abstract not $t : \Phi$)

Propositional equality

Formation

$$\frac{A \text{ type} \quad s:A \quad t:A}{s =_A t : \text{prop}}$$

$$\frac{t : A}{\text{refl}_t : t =_A t}$$

Intro

Elim:

$$\mathcal{J}_A : \Pi_{C:(\Pi_{x:A}\Pi_{y:A}((x =_A y) \to \text{Type}))} (\Pi_{x:A} C\, x\, x\, (\text{refl}_{A\,x})) \to \Pi_{x:A}\Pi_{y:A}\Pi_{p:(x =_A y)} C\, x\, y\, p$$

Computation:

$$\mathcal{J}_A\, C\, f\, t\, t\, \text{refl}_{A,t} \to f\, t$$

Equality type

sodbena enakost, ne nevaehne

$$\frac{u : B[s/x] \qquad s \equiv t : A}{u : B[t/s]}\ \text{substitution rule}$$

$$\frac{s : A \qquad t : A \qquad s \xrightarrow{*} u \qquad t \xrightarrow{*} u}{s \equiv t : A}$$

SAT-solving (propositional logic)

variable

$\Phi ::= x \mid \neg\Phi \mid \bot \mid \Phi \vee \Psi \mid \top \mid \Phi \wedge \Psi$

other conectives can be defined

We view formula Φ with at most k propositional variables $x_1, \dots, x_k$ as a function

$$[\![\Phi]\!] : \{0,1\}^k \longrightarrow \{0,1\}$$

semantic brackets

- Φ in variables $x_1, \dots, x_k$ is satisfiable if exists $\vec{b} \in \{0,1\}^k$, such that the formula is true on that assigment of bits
- formula is a tautology or valid if it's true for $\forall \vec{s} \in \{0,1\}^k$. $[\![\Phi]\!](\vec{s}) = 1$
- Φ and Ψ are ekwivalent if for $\forall b \in \{0,1\}^k$. $[\![\Phi]\!](\vec{b}) = [\![\Psi]\!](\vec{b})$

b is called satisfying assigment

Conections with the proof system

- Soundness theorem

if we can prove Φ (from no assumptions) in natural deduction, then Φ is valid.

- completeness theorem

if Φ is valid we can prove Φ in nat. deduction

(we need raa or low of excluded middle)

Φ is valid $\Leftrightarrow$ $\neg\Phi$ is not satisfiable

Definitions give rise to "truth table" algorithms, that check Φ for all $\vec{b} \in \{0,1\}^k$

Proof system prove alternative algorithm for validity: search for a proof

Negation normal form

Φ is in neg. normal form (NNF) if the only negations appear directly in front of propositional variables

$$\Phi ::= x \mid \neg x \mid \bot \mid \Phi \vee \Psi \mid \top \mid \Phi \wedge \Psi$$

Algorithm for conversion to nnf

$$\neg(\Phi \vee \Psi) \longrightarrow (\neg\Phi) \wedge (\neg\Psi)$$
$$\neg(\Phi \wedge \Psi) \longrightarrow \neg\Phi \vee \neg\Psi$$
$$\neg\neg\Phi \longrightarrow \Phi$$
$$\neg\bot \longrightarrow \top$$
$$\neg\top \longrightarrow \bot$$

(nondeterministic rewrite algorithm)

Theorem:

For any formula Φ, the rewrite algorithm terminates with the formula Φ^{nnf} that is equivalent to Φ.

Time complexity is $O(|\Phi|)$ ← size of Φ

also $|\Phi^{nnf}| = O(|\Phi|)$

A literal is a prop. variable x or the negation of x

A clause is a finite disjunction of literals

$L_1 \vee \ldots \vee L_n$ or $\bot$

(A coclause in a finite conjunction of literals

$L_1 \wedge \ldots \wedge L_n$ or $\top$)

A formula is in conjunctive normal form if it is a finite conjunction of clauses

$D_1 \wedge \ldots \wedge D_n$ or $\top$ wher D_i is a clause

A formula is in disjunctive normal form

$C_1 \vee \ldots \vee C_n$

One can give an algorithm that converts any formula Φ to Φ^{cnf} (notes)

or to Φ^{dnf}

The SAT problem

Input: is a prop. formula Φ
Output: yes if Φ is satisiable and
no if Φ is not satisfiable

The useful SAT problem

Input: is a prop. formula Φ
Output: yes if Φ satisfiable + a satisfining assignment,
no if not satisfiable

Famously SAT is NP-complete

There is no known efficient in theory algorithm

The DNF SAT problem

Input: formula in DNF
Output: yes or no

There is an easy algorithm in $O(n^2)$

The CNF sat problem (the useful one)
input: formula in CNF
output: yes or no

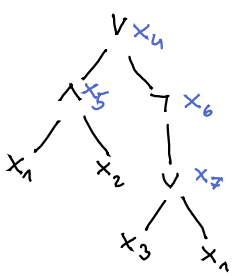
The Tseytin transformation

maps $\Phi(x_1 \dots x_k)$ to $\Phi^T(x_1 \dots x_K)$ where $K \geq k$ such that

- Φ^T is 3-CNF (each contains at most 3 literals)
- $(b_1, \dots b_K) \mapsto (b_1 \dots b_k) : \{0,1\}^K \longrightarrow \{0,1\}^k$ is a bijection satisfying assignments for Φ^T and satisfying assignments
- Φ^T is complete in $O(|\Phi|)$ time

Example

Φ: $(x_1 \wedge x_2) \vee \neg(x_3 \vee x_1)$



$x_4 \Leftrightarrow x_5 \vee x_6$

$x_5 \Leftrightarrow x_1 \wedge x_2$

$x_6 \Leftrightarrow \neg x_7$

$x_7 \Leftrightarrow x_3 \vee x_1$

implikacija $\Rightarrow$ dve implikaciji $\Leftarrow$

1.) $(\neg x_4 \vee x_5 \vee x_6) \wedge (\neg x_5 \vee x_4) \wedge (\neg x_6 \vee x_4)$

2.) $(\neg x_5 \vee x_1) \wedge (\neg x_5 \vee x_2) \wedge (\neg x_1 \vee \neg x_2 \vee x_5)$

3.) $(\neg x_6 \vee \neg x_7) \wedge (x_7 \vee x_6)$

4) $(\neg x_7 \vee x_3 \vee x_1) \wedge (\neg x_3 \vee x_7) \wedge (\neg x_1 \vee x_7)$

5.) $x_4 =$ 1) $\wedge$ 2) $\wedge$ 3) $\wedge$ 4)

Example: 3-colorability as SAT problem

Let (V, E) be a graph (undirected, irreflexive)

Construct a formula Φ_G, with

prop variables $\{r_v, g_v, b_v \mid v \in V\}$

Φ_G in constituded as CNF formula

as a conjunction of clauses

for every $v \in V$ we include

1.) $r_v \vee g_v \vee b_v$

2) $(\neg r_v \vee \neg g_v) \wedge (\neg g_v \vee \neg b_v) \wedge (\neg r_v \vee \neg b_v)$

(vsak v ima natanko eno barvo

for every (v, v') in E we include

3) $\neg r_v \vee \neg r_{v'}$ $\neg g_v \vee \neg g_{v'}$ $\neg b_v \vee \neg b_{v'}$

Satisfying assignment for Φ_G are in 1:1 correspondence with 3-coloring of G

DPLL (Φ, ρ) ρ assignmet of truth variable

Repeat until is aplicable

- if Φ is [] return ρ as the satisfying assignment
- if Φ containes ()
- if Φ conteines a unit clause L: $\rho' = \rho, L$, DPLL(simp(Φ, L), ρ')

if none of the above aply, we chose som literal L in Φ and continue with $\rho' := \rho, L$

$\Phi' =$ simplify (Φ, L). continue DPLL (Φ', ρ')

if DPLL returns ρ'' then return ρ''

otherwise DPLL $\rho, \neg L$ and $\Phi' =$ simplify$(\Phi, \neg L)$

$[(p \vee q) \wedge (\neg p \vee r) \wedge (\neg g \vee \neg r) \wedge (\neg p \vee g \vee \neg r)]$

choose p

$[r \wedge (\neg g \vee \neg r) \wedge (g \vee \neg r)$ $\qquad p$

unit close r

$[\neg g \wedge g]$ $\qquad p, r$

unit close $\neg g$

$[()]$ $\qquad p, r, \neg g$

consider $\neg p$

$[g \wedge (\neg g \vee \neg r)]$ $\qquad \neg p$

unit clause g

$[\neg r]$ $\qquad \neg p, g$

unti clause $\neg r$

$[]$ $\qquad \neg p, g, \neg r$

Algorithmic problems in predicate logic

In predicate logic we have a signature: a set $\mathcal{P}$ of relation symbols or predicates, a set $\mathcal{F}$ of function symbols

P has arryti ≥ 0

f has arryti ≥ 0

formulas

$$\Phi ::= P(t, \ldots, t) \mid t = t \mid \neg\Phi \mid \Phi \vee \psi \mid \Phi \wedge \psi \mid \forall x. \Phi \mid \exists x. \Phi$$

$$t ::= x \mid f(t, \ldots t)$$ terms

Example signature $(\leq, 0, +)$

constant

relation arryti 2

Example formula: $\forall x.\ 0 \leq x$

is true in natural interpretation on $\mathbb{N}$

it is false in $\mathbb{Z}$ and other interpretations in $\mathbb{N}$

A sentence is a formula with no free variables

Structures (Models)

A structure $\mathcal{M} = (M, (P^{\mathcal{M}})_{P \in \mathcal{P}}, (f^{\mathcal{M}})_{f \in \mathcal{F}}$

where M is set (domain or universe of quantification)

for every relation symbol P of arrity n

$$P^{\mathcal{M}} : M^n \longrightarrow \{\top, \bot\} = \{t, f\}$$

for every function symbol f of arrity n

$$f^{\mathcal{M}} : M^n \longrightarrow M$$

Example structure

$\mathcal{L}$

$L := \{0,1\}^*$

set of finite sequences

$0^{\mathcal{L}} := \varepsilon$ empty sequence

$+ := w_1, w_2 \mapsto w_1 w_2$ (concatenation)

$\leq := w_1, w_2 \mapsto \begin{cases} \top & \text{if } w_1 \text{ is prefix of } w_2 \\ \bot & \text{otherwise} \end{cases}$

The sentence

natural interpretation of $\mathbb{N}$

$\forall x \forall y \; x+y = y+x$ holds in $\mathcal{N}$, but not in $\mathcal{L}$

We write $\mathcal{N} \models \forall x \forall y \; x+y = y+x$

$\mathcal{L} \not\models \forall x \forall y \; x+y = y+x$

$\mathcal{M} \models \Phi$ .... $\mathcal{M}$ satisfies Φ

The satisfaction relation

$\mathcal{M} \models_\rho \Phi$

environment

(function mapping variables to elements M)

$$\boxed{\begin{array}{l} \mathcal{L} \models x \leq y \\ \quad x \mapsto 01 \\ \quad y \mapsto 011 \\ \\ \mathcal{L} \not\models x \leq y \\ \quad x \mapsto 000 \\ \quad y \mapsto 0101 \end{array}}$$

$$\left| \begin{array}{l} x^\rho := \rho(x) \\ (f(t_1 \ldots t_n))^\rho := f^{\mathcal{M}}(t_1^\rho, \ldots t_n^\rho) \end{array} \right.$$

$$\mathcal{M} \models_\rho P(t_1 \ldots t_n) \Leftrightarrow P^{\mathcal{M}}(t_1^\rho, \ldots t_n^\rho) = +$$

$$\mathcal{M} \models_\rho t_1 = t_2 \Leftrightarrow t_1^\rho = t_2^\rho$$

$$\mathcal{M} \models_\rho \Phi_1 \wedge \Phi_2 \Leftrightarrow (\mathcal{M} \models \Phi_1) \wedge (\mathcal{M} \models \Phi_2)$$

$$\mathcal{M} \models_\rho \forall x\, \Phi \Leftrightarrow \text{for all } a \in M,\ \mathcal{M} \models_{\rho[x \mapsto a]} \Phi$$

Huth and Ryan textbook

A sentence Φ is valid if for all structures $\mathcal{M}$
$\mathcal{M} \models \Phi$

Soundness theorem:

if sentence Φ has a proof in natural deduction, then Φ is valid

Completness theorem:

if Φ is valid, then it has proof in ND.

Φ is satisfiable if there exist a structure, such that $\mathcal{M} \models \Phi$

The validity problem:

input: a sentence Φ

output: yes if Φ valid, no otherwise

Theorem (Church/Turing)

The validity problem is undecidable

A problem is decidable if there is an algorithm for it, that is garentied to terminate with the correct answer

it follows that the satisfiability problem is also undecidable

A practical algorithmic problem:

input: a sentence Φ and a puprorted ND proof of Φ in ND

output: yes if proof is correct, no otherwise

This is efficently solvable

(used in proof assistance)

Consider following algorithm for validity

Algorithm (Φ)

systematicly search the space of all proofs
until we find a proof of Φ
if we find a proof, return yes
it never returns no :(

This algorithm shows that validity is semidecidable (A yes no is semidecidable if there exist an algorithm that always correctly answers yes, if the answer is yes but runs forever otherwise)

Satisfiability is cosemi decidable

cosemi decidable means that it answers no if false, otherwise runs forever

Modeling a computer network

Signature: Connected /2 ; server, client /1

$\forall x \ \neg \text{Connected}(x,x)$

$\forall x,y. \ \text{connected}(x,y) \rightarrow \text{connected}(y,x)$

$\forall x. \ \text{server}(x) \vee \text{server}(x)$

$\forall x. \ \neg \text{server}(x) \vee \neg \text{client}(x)$

$\forall x. \ \text{client}(x) \rightarrow \exists y. \ \text{server}(y) \wedge \text{connected}(x,y)$

$\forall x. \ \text{server}(x) \rightarrow \exists y. \ \text{client}(y) \wedge \text{connected}(x,y)$

$\forall x,y. \ \text{connected}(x,y) \rightarrow \neg \text{server}(x) \vee \neg \text{serve}$

possible consider to have network with at least 10 devices ~~servers~~ ctu that exactly 2 are ser

$$\exists x_1, \dots x_{10}. \ \bigwedge_{i=1}^{9} \bigwedge_{j=i+1}^{10} x_i \neq x_j$$

$$\wedge \ \exists x \exists y \ x \neq y \wedge \forall z. \ \text{server}(z) \rightarrow z = x \vee z = y)$$

$$\wedge \ \text{server}(x) \wedge \text{server}(y)$$

Finite satisfiability

Question: sentence Φ

Answer yes if exists finite structure $\mathfrak{A}$, such that $\mathfrak{A} \models \Phi$ and no otherwise

Finite satisfiability is semidecidable

Algorithm: for every size $n = 1, 2, \ldots$ we enumerate all non-isomorphic structures of size n and for each structure $\mathfrak{A}$ we look at the model checking problem $\mathfrak{A} \models \Phi$, if and when we find such a structure, we return that structure

24.3 trisum ~~pi~~ pisela

SMT-solving

31.3

Satisfy modulo a theory

SAT	prop. logic	not very expressive but solving is practical
FOL	...	good expressivity but solving is impractical (but not entirely)
SMT	the swet spot in the middle. Expressive enough and satisfiability is still practical	

SMT basic setting:

First order signature: function symbols + predicates

Tipicly we are interested in the question:

Is Φ satisfiable modulo a theory T ?

For our purposes a theory T is determined by a set of structures mod_T

We are interested in the question

$$\exists \mathcal{M} \in Mod_T. \exists \text{environment } \rho. \mathcal{M} \models_\rho \Phi \ ?$$

SMT is useful for program verification.
Construct $\phi(x_1, \dots, x_n)$ saying $x_1, \dots, x_n$ are inputs on which the program behaves incorrectly.
A satisfying assignment to ϕ finds a bug in input.
If unsatisfiable, there are no bugs

Interesting theories:

EUF: equality between uninterpreted functions

signature: just function symbols

Flatten formula: no nesting of function symbols, every term named by a variable

Perform congruence closure

Check consistently of the resulting congruence closure with negated equality

$$x=y \wedge x=g(y) \wedge u=f(x,g(x)) \wedge v=f(y,x) \wedge u \neq v$$

$$\underbrace{x=y} \wedge \underbrace{x=g(y) \wedge t=g(x)} \wedge u=f(x,t) \wedge v=f(y,x) \wedge \underbrace{u \neq v}$$

$[x|y|t|u|v]$ ... partition of all variables into separate equivalence classes

$[x,y|t|u|v]$

$[x,y,t|u|v]$

$[x,y,t|u,v]$ $\Rightarrow$ formula is unsatisfiable

- Have algorithm congruence closure decide satisfiability for conclauses (conjunctions of literals) (practical)
- So we can decide satisfiability arbitrary quantifier free formulas by reduction to DNF (impractical)

Anja Petković

Theorry of arrays:

Two functions : $write(a, i, v)$, $read(a, i)$

Theory: $i = j \rightarrow read(a, i) = read(a, j)$

$i = j \rightarrow read(write(a, i, v), j) = v$

$i \neq j \rightarrow read(write(a, i, v), j) = read(a, j)$

Reducing array to EUF

$i \neq j \wedge u = read(write(a, i, v), j) \wedge v = read(a, j) \wedge u \neq v$

$i \neq j \wedge u = (i = j \; ? \; v : read(a, j)) \wedge v = read(a, j) \wedge u \neq v$

ite pdtm sicer

$i \neq j \wedge u = read(a, j) \wedge v = read(a, j) \wedge u \neq v$

$[i | j | u | v]$ Now we perform satisfiability test

$[i | j | u, v]$ unsatisfiable!

What we aim for with SMT?

We have a theory, that supports efficent satisfiability for cocluses. We want to bootstrap to arbitrary quantifier free formulas

2 approaches:

- Eager: Translate to a propositional sat problem
- Lazy: Modify SAT solving algorithm based on DPLL. To find contradiction we use coclause decision procedure

start with a flatten formula

$$x=f(y) \wedge y=f(x) \wedge x=y$$

replace terms with variables

$$x=f^y \wedge y=f^x \wedge x=y \wedge (x=y \rightarrow f^x=f^y)$$

We obtain in the end a first order formula, where all atomic formulas are equalities between variables

Let k be the smallest number, such that $2^k \geq$ number of variables.

Bit blasting: replace every variable x with k propositional variables $x_1, \dots, x_k$

Replace every equality $x=y$ with

$$x_1 \leftrightarrow y_1 \wedge x_2 \leftrightarrow y_2 \wedge \dots \wedge x_k \leftrightarrow y_k$$

$x \neq y$ with $\neg(\dots)$

we submit to SAT solver.

This will be satisfiable $\Leftrightarrow$ it's unsatisfiable

start with flatten formula $x=f(y)$, $y=f(x)$, $x \neq y$

$$x=f^y \wedge y=f^x \wedge x \neq y \wedge (x=y \rightarrow f^x=f^y)$$

$k=2$

$$(x_1 \leftrightarrow f_1^y) \wedge (x_2 \leftrightarrow f_2^y) \wedge (y_1 \leftrightarrow f_1^x) \wedge (y_2 \leftrightarrow f_2^x) \wedge$$

$$\neg(x_1 \leftrightarrow y_1 \wedge x_2 \leftrightarrow y_2) \wedge ((x_1 \leftrightarrow y_1 \wedge x_2 \leftrightarrow y_2) \rightarrow$$

$$f_1^x \leftrightarrow f_1^y \wedge f_2^x \leftrightarrow f_2^y$$

Linear arithmetic

Terms $0 \quad + \quad 1 \quad a \cdot x$

integer/natural/real/rational ....

$\leq$

Model Checking

Today: Linear Time Temporal Logic (LTL)

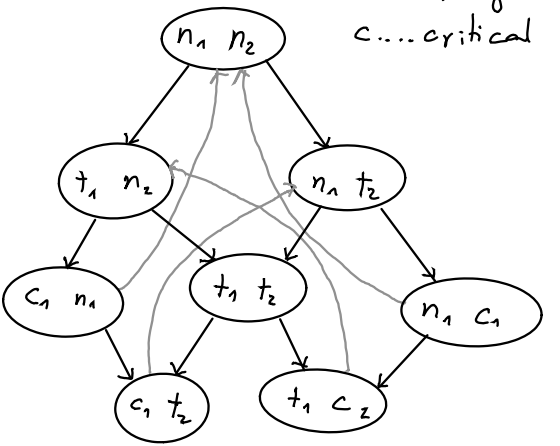
Further reading: HR, chapter 3

Example of transition system:

n ... nonctritical

t ... trying

c ... critical



Example safety property (the system always stays in safe state

globalno ← 1 in 2 nista critical na enkrat

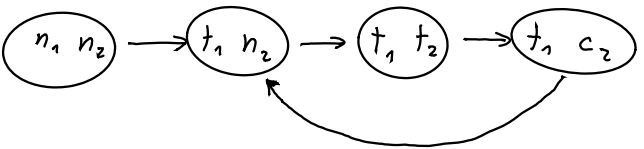
$G(\neg(c_1 \wedge c_2))$ trivialy holds

We know also liveness property (something desirable will eventually happen)

$$G((t_1 \rightarrow F\, c_1) \wedge (t_2 \rightarrow F\, c_2))$$

(če 1 poskuša dobiti dostop, ga bo enkrat dobila)

This does not hold. Counter run:



This is called „lasso" run

A transition system is given by

$(S, \rightarrow, L, Atoms)$

where S : s finite set (of states)

$\rightarrow \subseteq S \times S$ (the transition relation)

$L : S \rightarrow \mathcal{P}(Atoms)$ maps state to it's atoms

The labeling function

powerset of atoms

...

$Atoms = \{n_1, n_2, t_1, t_2, c_1, c_2\}$

now and in the future

Φ holds until Ψ holds

Φ holds globally

LTL formulas:

$$\Phi ::= \underset{\substack{\uparrow \\ \text{Atoms}}}{p} \mid \neg\Phi \mid \Phi \wedge \Phi \mid \Phi \vee \Phi \mid X\Phi \mid F\Phi \mid G\Phi \mid \Phi \, U \, \Psi$$

Φ holds in next

Φ holds in future

in textbook also $\mid \Phi \, R \, \Psi \mid \Phi \, W \, \Psi$

Requements for LTL

$\forall s \in S. \exists s' \in S. \; s \rightarrow s'$

The satisfaction relation take the form

$$\pi \models \phi$$

Here π is infinite run trought transition systems

$$\pi = s_1 \to s_2 \to s_3 \to \dots$$

satisfaction relation

$$\pi \models p \iff p \in L(s_1) \quad (p \in L(\pi(1)))$$

$$\pi \models \neg\phi \iff \pi \not\models \phi$$

$$\pi \models \phi \wedge \psi \iff \pi \models \phi \wedge \pi \models \psi$$

$$\pi \models X\phi \iff \pi^2 \models \phi$$

ϕ holds in run started on second

$$\pi \models G\phi \iff \forall n. \pi^n \models \phi$$

$$\pi \models F\phi \iff \exists n. \pi^n \models \phi$$

$$\pi \models \phi \mathcal{U} \psi \iff \exists n. \pi^n \models \psi \wedge \forall m < n. \pi^m \models \phi$$

$$\pi^n := s_n \to s_{n+1} \to \dots$$

Model checking problem

Input: A transition system $\mathcal{M}$

A system state s_0

An LTL formula Φ

Output:

yes, if $\forall$ runs π through $\mathcal{M}$ starting at s_0, we have $\pi \models \Phi$ $\quad m, s_0 \models \Phi$

No, since $\exists$ an infinite run π that fa. log

The livenes property for mutual exclusion

$$G((t_1 \rightarrow F c_1) \wedge t_2 \rightarrow F c_2))$$

DM: modify the model where this holds

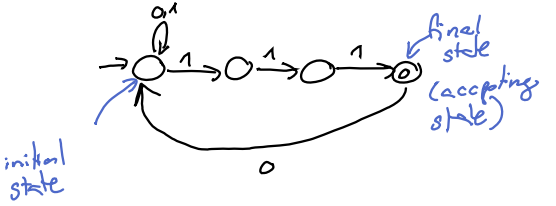
Alternative: Assume fairness property

A fairness assumption:

$$\Phi_{fairns} = GF(t_1 \wedge t_2) \rightarrow GF(c_1) \wedge GF(c_2$$

Model checking formula: $\Phi_{fairnes} \rightarrow \Phi$ liviences

Nondeterministic Büchi automata



Alphabet: $\{0,1\}$

A büchi automata defines a language of infinite words

The set of infinite words is written Σ^ω

Examples of ω-words

$000\ldots$

$0101\ldots$

$\ldots$

infinite word is accepted, if there is some run of the automata, labeled by the word, that passes accepting state infinetly times

Model checking problem for LTL

input: Transition system $\mathcal{M}$, starting state s, LTL formula ϕ

Output: yes, if $\mathcal{M}, s \models \phi$,
no, otherwise + lasso run π such that $\pi \models \neg\phi$

<u>Language theory</u> alphabet Σ, word is a finite sequence from Σ (Σ^* for the set of words)

A <u>langue</u> is a set of words, $L \subseteq \Sigma^*$

<u>Language theory for ω-words</u>

ω-word is an infinite segence from Σ
(Σ^ω for the set of ω-words)

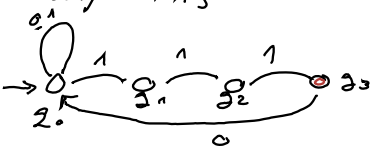
An <u>ω-language</u> is a set of ω-words $L \subseteq \Sigma^\omega$

An automate on ω-words specifies an ω-language

Büchi automata is a particular automata on ω - words

$\Sigma = \{0,1\}$ A_1 will specify:

$L_\omega(A_1) = \{ w \in \{0,1\}^\omega ;$ w contains substring 1110 infinetly often $\}$



A_2:

$L_\omega(A_2) = \{ w \in \{0,1\}^\omega \mid$ from some point ther are only ones. There are finitly many zeros)

A_3:

$L_\omega(A_3) = \{\}$

A Büchi automata is formelly a tuple

(Q, I, Δ, F) A

Q: set of states

I: set of initial states $I \subseteq Q$

F: set of final states $\Delta \subseteq Q$

Δ: Transition relation $\Delta \subseteq Q \times \Sigma \times Q$

nondeterministic büchi automata

Proposition (Emptyness testing for an NBA)

Let A be an NBA

$L_\omega(A) \neq \emptyset \iff$ there exists a path of the form

$$q_0 \to q_1 \to \cdots q_n \to \cdots$$

initial state

final state

Such a path is called an accepted lasso

Definition: An ω-language $L \subseteq \Sigma^\omega$ is ω-regular if there exists NBA A such that $L_\omega(A) = L$

Theorem: ω-regular languages are closed

- under union
- intersection
- complement (Büchi's theorem)

Consider an CTL formula $a \mathcal{U} b$

Consider when does it hold that a run satisfies

$\pi \models a \mathcal{U} b$

$\pi = s_1 \to s_2 \to s_3 \to \dots$

The Trace of a run $\pi = s_1 \to s_2 \to \dots$

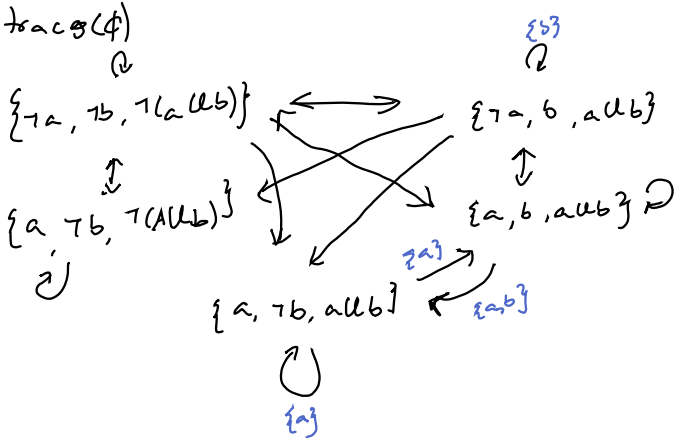
is the sequence $L(s_1), L(s_2), L(s_3) \dots$ ← label function

This is an ω-word in $\mathcal{P}(\text{Atoms})^\omega$

The truth of a formula ϕ on a run π is completly determined on the trace

$\text{Traces}(\phi) := \{ w \in \mathcal{P}(\text{Atoms})^\omega ;$ ϕ is true on an any run with $w \}$

Theorem: For any ϕ, Traces(ϕ) is ω-regular.
There is an algorithm to construct A_ϕ recognising traces(ϕ)

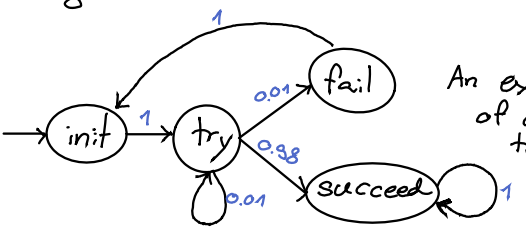


Model Checking algorithm
step 1 construct the automaton $A_{\neg\phi}$
step 2: product $A_{\neg\phi} \otimes \mathcal{M}$

Probabilistic Model Checking

In general:

- many probabilistic models (today we focus on DTMCs ← descrete time Markov chain
- many probabilistic logics (PCTL, PCTL*) (we arg going to use LTL)
- There is a whole technology of associated algorithems



An example of descrete time Markov chain

trying until you succeed (device is not available

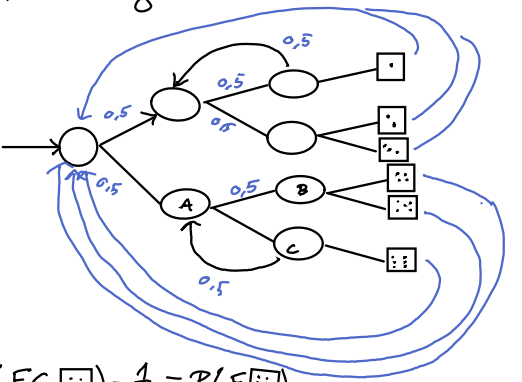
$P(init) = 1 \qquad P(X\, try) = 1 \quad P(XX\, succ) = 0.98$

$P(X(try \wedge X(try \wedge X\, try))) =$

$= P(XX(try \wedge X\, try)) = P(XXX\, try) = 0.0001$

$P(FG\, succ) = 1 - \underbrace{(0{,}01)^n}_{\tau_{try}} \underbrace{(0{,}01)^m}_{\tau_{fail}} \xrightarrow{n+m \to \infty} 1$

Knuth/Yao-algorithem for a fair dice



$$P(FG\,\boxed{\cdot\cdot\cdot}) = \frac{1}{6} = P(F\,\boxed{\cdot\cdot\cdot})$$

modified: $P(O\,\mathcal{U}\,\boxed{\cdot\cdot\cdot}) = \frac{1}{6}$

$$P(O\,\mathcal{U}(\boxed{\cdot\cdot\cdot} \wedge X(O\,\mathcal{U}\,\boxed{\cdot\cdot\cdot})) = \frac{1}{36}$$

$P_B = 0$ $\quad P_C = \frac{1}{2} + \frac{1}{2} P_A$ $\quad P_A = \frac{1}{2} P_B + \frac{1}{2} P_C =$

$= \frac{1}{4} + \frac{1}{4} P_A \rightsquigarrow P_A = \frac{1}{3}$

General solution:

We have DTMC and LTL formula Φ

Question: What is the probability that a randomly generated run π satisfies Φ

Two issues:

1. Make sense of the question
2. Compute the answer

A DTMC is (S, s_{init}, P, L)

transition matrix

set of states

labeling function

- S set of states (in modeling is normaly finite)
- $s_{init} \in S$ the initial state
- L labeling function $: S \to \mathcal{P}(Atoms)$
- $P : S \times S \to [0,1]$ transition probability matrix

  satisfying $\forall s \in S. \sum_{t \in S} P(s,t) = 1$

for a formula Φ we want to calculate

$$\mathbb{P}[\Phi] := \mathcal{P}(\{\pi \in Runs \mid \pi \models \Phi\}$$

We assign probability to mesurable sets of runs

A cylinder set is determined by a finite run

$w = s_1 \xrightarrow{p_1} s_2 \xrightarrow{p_2} \dots \to s_n \qquad s_1 = s_{init} \qquad p_i > 0$

$$\langle w \rangle := \{\pi \in Runs \mid \pi \text{ extends } w \ (w \text{ is prefix of } \pi)\}$$

$$\mathbb{P}\langle w \rangle = \prod_{i=1}^{n-1} p_i$$

The mesurable subset of Runs are defined by

- Every cylinder set $\langle w \rangle$ is mesurable
- If $R_1, R_2, \dots$ are mesurable sets then so are $\bigcup_n R_n$, $\bigcap_n R_n$
- if R is mesurable, than so is R^c
- $\emptyset$ and Runs are mesurable

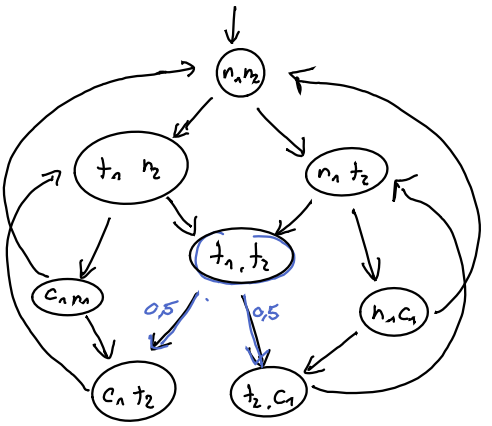
A general fact: Let $\mathcal{E}$ for the set of all mesurable subset of Runs. There is a unique probability function $\mathbb{P}: \mathcal{E} \to [0,1]$ satisfying

$\mathbb{P}$ is a probability measure
- $\mathbb{P}\emptyset = 0$
- $\mathbb{P}\langle w \rangle$ is as defined earlier
- If we have $R_1, R_2, \ldots$ pairwise disjoint mesurable sets then $\mathbb{P}(\bigcup_n R_n) = \sum_n \mathbb{P} R_n$ (countable additivity

Proposition: For any formula Φ, the set $\{\pi \in Runs \mid \pi \models \Phi\}$ is mesurable.

The proof uses Büchi automata

Model checking: Given (S, s_{init}, P, L) and Φ answer $P_{>g}\Phi$ = is $\mathbb{P}(\Phi) > g$



This is a Markov decision process (MDP) is combining probability and determinism

Liveness: $G(t_1 \to F c_1) \wedge (t_2 \to F c_2)$

A Markov run would be

$(n_1, n_2) \to n(t_1) \xrightarrow{0,5} (c_1, t_2) \to (n_1, t_2) - \ldots$

$n(t_1) \xrightarrow{0,5} (c_2, t_1) \to \bigcirc - \ldots$

livness: $G(t_1 \to F c_1) \wedge (t_2 \to F c_2)$ has probability 1 over all markov chains